# TYPESPRINT

Complete Typing, Gaming and Learning Platform

MASTER DEVELOPMENT PLAN | v1.0 | 30 October 2026

**Development workflow:** Codex first; Google Antigravity as fallback.

**Monetization:** Free product with Google AdSense after eligibility and approval.

# 1. Executive summary and scope

TypeSprint is a responsive, browser-first typing platform combining four top-level areas: (1) YouTube Type Master, (2) Typing Games, (3) General Typing, and (4) Beginner Typing Academy. Typing Games contains Ninja Battle, Shadow Battle, offline Car Racing, and online multiplayer Car Racing. The interface should look and feel like a polished gaming product while preserving real educational value.

**Core principle:** One independently testable TypeScript typing engine drives every experience. Balanced penalties are the default. Free access and organic search are the business model; advertising income is neither immediate nor guaranteed.

## Goals

• Deliver smooth, accurate typing input with clear WPM, accuracy, streak and error statistics.

• Make offline modes instantly usable without login or a paid third-party service.

• Offer genuine beginner progression and entertaining game loops that encourage repeat sessions.

• Release a useful public beta early; build expensive multiplayer only after core validation.

## Out of scope for initial release

• AI language-model enemies, complex multiplayer tournaments, 3D open worlds, mobile native apps, copyrighted lyrics/transcripts, intrusive ad mechanics.

# 2. Information architecture and user journeys

| Top-level area | Main activities | Initial account requirement |
|---|---|---|
| YouTube Type Master | Paste eligible URL; watch visible embedded video; type original scrolling exercises; view results | Guest |
| Typing Games | Ninja Battle, Shadow Battle, offline and online Car Racing | Guest for offline; account for ranked online |
| General Typing | Timed speed tests, accuracy mode, numbers, punctuation, daily practice | Guest |
| Beginner Academy | Guided keyboard lessons, feedback, checkpoints, progress | Guest local; account for cloud sync |

**Primary navigation:** Home | YouTube Type Master | Typing Games | General Typing | Beginner Academy | Leaderboards | Profile. Home features four highly visible cards, quick-start general test, beginner entry point and game previews.

# 3. Shared typing engine: definitive rules

## Input and time

• A session begins on its first permitted real keyboard input. Monotonic elapsed time is the scoring source; animation frames must never determine time.

• Track expected text, raw keystrokes, correct keystrokes, mistakes, corrected errors, uncorrected errors, completed words, streak, session state, focus changes and elapsed duration.

• Ignore or block paste in scored/ranked challenges; handle key repeat, composition/IME input, Backspace, modifiers, browser shortcuts and accessibility.

• Allow pause/restart in practice and offline modes. In ranked multiplayer, pausing cannot freeze the server-side race.

### Default performance metrics

**Gross WPM** = (characters entered / 5) / elapsed minutes. **Accuracy** = correct keystrokes / total scored keystrokes * 100. **Effective WPM** = gross WPM * accuracy fraction. Final implementation must document precisely which keys count and remain consistent across UI and saved results.

### Balanced penalty contract

• Wrong inputs receive red visual feedback and reduce combo quality / acceleration / attack power; a single error does not end a session.

• Correct successive inputs restore the streak and normal effectiveness. Consecutive mistakes apply stronger but capped penalties.

• Strict error-blocking can be an optional practice setting; arcade scoring can be an unranked alternate later.

• Provide game-specific adapters: CombatAdapter, RacingAdapter, MusicChallengeAdapter, AcademyAdapter. The core engine never directly manipulates game visuals.

## 4. YouTube Type Master

The user pastes a YouTube URL; the app validates supported link formats, resolves video metadata with the official API, checks embeddability where possible, and rejects videos longer than 05:00. Private, deleted, age-restricted or non-embeddable items require a clear fallback error. API keys must remain server-side with quotas and abuse safeguards.

### YouTube Companion Mode (MVP)

• Keep the official YouTube embed player visible and usable with standard controls. The user explicitly starts playback and typing.

• TypeSprint scrolls **original practice text** independently alongside the video. Highlight current word, correct letters, errors, progress, combos, and completion.

• At 5 seconds without typing, pause the **typing challenge** and show an idle indicator; do not rely on manipulating YouTube playback for penalties.

• Mistakes reduce typing score and text-flow rewards; they do not force video seeking, skipping, stopping, or changing playback.

• If video ends first, offer to finish the challenge or see partial results. Manage player initialization, playback errors and resuming the separate challenge.

### Optional original-audio Rhythm Mode (later)

• For audio owned or properly licensed by TypeSprint, implement precise start/pause/resume, word-beat timing, and configurable error skips. Separate this from YouTube integration.

**Policy check before implementation:** Verify the latest YouTube API Terms, embedded-player requirements and developer policies. Avoid downloading videos, obtaining copyrighted lyrics/transcripts, or hiding/overriding the YouTube player.

## 5. Ninja Battle specification

2D side-view action game: a player ninja faces one state-machine AI ninja. Correct word completion initiates attacks; accuracy scales damage, consecutive words build combos, and a special meter unlocks a high-impact move. Characters need idle, run, attack, hit, block, victory and defeat animations plus responsive audio and reduced-motion settings.

| Setting | Easy | Medium | Hard |
|---|---|---|---|
| AI attack interval | ~7 sec | ~5 sec | ~3 sec |
| AI damage | 5 HP | 8 HP | 11 HP |
| Target length | 3-5 letters | 5-8 letters | 7-12 letters |
| Mistake damage penalty | 5% | 10% | 15% |

• One round ends on 0 HP. Include health indicators, target word, typed feedback, combo meter, special meter, countdown, pause for offline play, and immediate rematch.

• Define enemy states: idle, approach, windup, attack, recovery, defend, hitstun, defeated. Timers and animation must clean up after game reset.

## 6. Shadow Battle specification

Shadow Battle must offer more than a recolored Ninja mode. Use a distinct dark environment, shadow-energy system, tactical defensive windows, faster patterns and boss stages.

• **Shadow Slash:** successful short burst deals heavier damage; **Dark Shield:** completing a defensive sequence blocks an incoming hit; **Phantom Dash:** timed input avoids an attack.

• Easy teaches mechanics; Medium introduces counters and timing; Hard introduces multi-stage patterns, complex punctuation and bosses.

• Each boss telegraphs attacks clearly; penalties never create unavoidable damage. Validate health, ability cooldown, combo reset and restart across repeated runs.

## 7. Car Racing - offline

A visible track must scroll continuously with responsive car movement, scenery, distance traveled, position indicators, checkpoints and a finish line. Correct typing advances the car via acceleration, high accuracy improves efficiency, and incorrect inputs cause temporary acceleration penalties. Idle typing results in gradual deceleration instead of an instant stop.

| Difficulty | AI target pace | Behaviour |
|---|---|---|
| Easy | ~20 WPM | Visible pauses and variable pace |
| Medium | ~40 WPM | Consistent speed, occasional errors |
| Hard | ~65 WPM | Strong and stable pace |

• Use a deterministic or seeded AI driver with bounded variance, clear disclosure that the opponent is AI, and repeatable test fixtures.

• One polished starter course, one AI opponent, countdown, ~60-120 second races, finish line, win/lose state, results and replay.

• Decide race movement from accepted typing events (not raw key spam). Avoid farming infinite progress by typing unrelated keys.

## 8. Car Racing - online multiplayer

Start with private 1v1 rooms, then add public matchmaking. Both players receive identical normalized text, synchronized server start, authoritative race progress, final results and disconnect handling. Online requires a separate reliable real-time service; do not assume serverless HTTP functions can maintain persistent WebSocket rooms.

• Lobby: create/join short-lived room code, readiness, matchmaking timeout and cancel. Never use a demo room as a real connection.

• Server: validates incremental input sequence, timing plausibility, accepted character progress, match state and winner. Rate-limit events and protect against client-fabricated WPM.

• Fair play: handle packet loss, reconnect grace, disconnect winner, server clock, match expiry, spectators later, and ranked integrity flags.

• Security: authenticated match participation, room membership checks, protection against replay, bounds on event size, profanity-safe display names.

• Acceptance: two real browser clients finish identical challenge; server and both clients agree on winner even when one disconnects/reconnects.

## 9. General Typing

• Timer presets: 15, 30, 60 and 120 seconds; mode variations: common words, sentences, accuracy, numeric entry, punctuation, custom practice and daily challenge.

• Results: gross and effective WPM, accuracy, correct/wrong keys, common problematic keys, speed-over-time graph, personal record and retry.

• Use original, sufficiently diverse and level-appropriate passages. Daily challenges must use consistent date and text selection.

• Save guest data to local storage with migrations/versioning; optionally synchronize user data to authenticated account later.

## 10. Beginner Typing Academy

| Stage | Learning focus | Suggested gate |
|---|---|---|
| 1 | Keyboard orientation and posture | Complete orientation |
| 2 | Home row ASDF JKL; | 90% accuracy |
| 3 | Upper/lower letter rows | 90% accuracy |
| 4 | Common words and spaces | 15 WPM |
| 5 | Shift and punctuation | 92% accuracy |
| 6 | Sentences | 25 WPM |

| Stage | Learning focus | Suggested gate |
|---|---|---|
| 7 | Speed/accuracy drills | 35 WPM practice |
| 8 | Graduation exercise | 30 WPM + 95% accuracy |

• Each lesson shows target key, finger indicator, visual keyboard, instructions, accuracy and progress. Detect frequently missed keys and suggest short remedial drills.

• Provide replay and skill advancement without confusing lockouts. User can resume from saved local progress and optionally sync accounts.

• Graduation certificate is a TypeSprint achievement, never represented as accredited employment certification.

## 11. UX and visual design requirements

• Use a cohesive premium dark theme with blue/cyan/purple accents, accessible contrast, optional light mode and consistent visual hierarchy.

• Separate four homepage experiences clearly; show a quick-start entry and preserve navigation without interrupting games.

• Desktop keyboard is the priority for timed modes; mobile supports navigation and education, with appropriate external-keyboard guidance.

• Animations must feel responsive, offer reduced motion and never block key events. Audio can be muted; respect browser autoplay restrictions.

• Ad placement cannot overlap typing controls, start/restart buttons, gameplay, or customer-critical interactions.

## 12. Architecture and tech stack

| Component | Proposed technology |
|---|---|
| Website | Next.js with TypeScript (App Router) |
| UI | Tailwind CSS; accessible UI components |
| Typing engine | Framework-independent TypeScript state machine |
| 2D games | Phaser 3 (validate current compatibility during setup) |
| Optional 3D | Three.js later, only if justified |
| Local progress | Browser storage, versioned |
| Auth and data | Supabase Auth + PostgreSQL with RLS, phased in |
| Multiplayer | Dedicated Node.js real-time authoritative server/WebSockets |
| YouTube | Official IFrame Player API + server-protected Data API |
| Testing | Unit, integration, Playwright browser end-to-end |
| SEO/analytics | Sitemap, metadata, Search Console, privacy-aware analytics |

| Component | Proposed technology |
| --- | --- |
| Hosting | Commercial-use-permitted deployment plan; separate multiplayer hosting |

### Code boundaries

Keep modules separate: app/routes, shared UI, typing-core, game adapters, game scenes, content, youtube-integration, persistence, multiplayer-server and tests. Favor typed event interfaces and deterministic simulation for tests. Never place secrets in public bundles.

### Suggested database entities

• profiles(user_id, display_name, settings); typing_sessions(user_id, mode, duration, stats, created_at); lesson_progress(user_id, lesson_id, best_score, completed_at).

• matches(id, mode, state, start_at, finish_at, seed, winner_id); match_players(match_id, player_id, verified_progress, status); achievements and leaderboard_entries.

• Apply user-specific row-level security, retention rules, input validation and data export/deletion controls.

## 13. Roadmap with verification gates

| Phase | Build | Exit gate |
| --- | --- | --- |
| 1 | Foundation, homepage, typing engine, General Typing | Correct tests, timer, input and scores |
| 2 | Beginner Academy | Eight stages, progress recovery, guidance |
| 3 | Ninja Battle | Complete win/lose/rematch loop, 3 levels |
| 4 | Shadow Battle | Distinct skills and boss gameplay |
| 5 | Offline Car Racing | Typing drives animated car; consistent AI results |
| 6 | YouTube Companion + optional Rhythm foundation | 5-minute validation and policy-safe experience |
| 7 | Accounts, private online 1v1, leaderboards | Two devices, validated results, disconnect tests |
| 8 | Public hardening, SEO, AdSense readiness | Accessible, fast, compliant, monitored |

**Release strategy:** Deploy a public beta after Phase 2, then add gameplay modes in stages. Phase 8 means full-platform stabilization rather than postponing all launch activity.

## 14. Acceptance and reliability test matrix

• **Typing:** correct/incorrect, Backspace, held keys, paste attempts, IME, input focus, browser tab switching, timed end and zero-duration handling.

• **Progress:** guest persistence, corrupted storage, replay, new user reset, lesson resume after refresh.

• **Combat:** all enemy difficulties, attack transitions, damage caps, death, special cooldowns, 10 consecutive restarts, cleanup without runaway timers.

• **Racing:** speed responds to valid typing, error penalty recovers, AI variation bounded, exact finish/winner calculation and replay.

• **Online:** two clients, network delay/loss, stale sessions, forged events, rematches, disconnect/reconnect and duplicated event IDs.

• **YouTube:** accepted/rejected formats, over-5-minute clip, missing metadata, disabled embed, rate limit, player errors and independent challenge idle.

• **Quality:** keyboard focus, screen reader labels, responsive widths, sound toggle, reduced motion, no unexpected console errors.

## 15. SEO, ads and business expectations

• Offer original educational pages and playable tools rather than thin keyword variants. Publish helpful typing lessons, descriptions, FAQ, About, Contact, Terms and Privacy pages.

• Add metadata, canonical URLs, robots and sitemap, structured data only where valid, Search Console and traffic measurements.

• Monetization is through Google AdSense after review and approval, with no guaranteed traffic, approval date, RPM, or income.

• No ads during active competitive play, intrusive overlays, or positions that invite accidental clicks. Do not encourage ad clicks.

• Verify current AdSense approval and YouTube API policies at integration time; provider terms can change.

## 16. Codex-first implementation operating procedure

• Create a Git repository. Keep this PDF in **docs/TypeSprint_Master_Development_Plan.pdf** as the product reference. Add README, decision log and a short phase checklist.

• Give Codex exactly one phase at a time, with scope exclusions, acceptance tests, and named deliverables. Avoid asking for the entire game in one prompt.

• Before editing, inspect only files relevant to the phase. Preserve working behaviour, avoid unrelated refactors and do not rewrite completed gameplay without explicit need.

• Require type checking, linting, unit tests, integration tests and a concise report of changed files, commands, results and remaining limitations.

• Commit after each validated milestone. If any regression occurs, compare with the last passing commit and fix narrowly.

• If Codex becomes unreliable, switch to Google Antigravity using the same PDF, latest Git commit, phase checklist, test results, known issues and exact unfinished scope. Do not mix unverified changes from both tools.

### Codex phase prompt contract

**Objective:** Implement Phase N only. **Inputs:** PDF specification, existing code and tests. **Non-goals:** all later phases. **Requirements:** named features and architecture boundaries. **Validation:** run tests and document manual checks. **Output:** changed files, screenshots if available, passing checks, known limitations and next-phase handoff.

## 17. Risks and mitigations

| Risk | Mitigation |
| --- | --- |
| Low search traffic | Publish useful lessons and differentiated games; test SEO and retention early |
| Complexity overload | Phase gates, share core engine, beta after Phase 2 |
| YouTube policy/embeds | Official player, independent typing challenge, original text |
| Competitive cheating | Server-authoritative matches and validation |
| High operating cost | Offline-first MVP, free/low-cost services where permitted |
| Low-end-device lag | 2D first, asset budgets, adaptive effects, event-driven typing |
| Ad approval denied | Build value without relying on ads; improve quality and compliance |
| Coding regressions | Git checkpoints, regression suites, narrow changes |

# 18. Definition of done

• All four homepage areas exist and navigate cleanly; all offline modes are fully playable; every session reports accurate statistics.

• Beginner path provides structured progression; Ninja and Shadow feel distinct; offline cars visibly race; YouTube mode complies with provider requirements.

• Online race passes two-device and integrity testing before enabling ranked play.

• The website deploys, functions on common modern browsers, has original learning content, and is ready for real users. AdSense revenue is a separate business milestone.

**Project status as of 30 October 2026:** Specification approved for planning; coding, assets, deployment, analytics and monetization are not yet implemented.